

Lab 5 Report

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1. An alternative to an encoder is a rotary potentiometer. What are the advantages that encoders have over rotary potentiometers? List at least two distinct advantages.

Encoder dont rely on a physical touch to sense. Encoders also have to ability to know the position they are in during power loss.

Topic: Question to think about

Based on the Pololu motor-encoder combo guide, what is the resolution of the quadrature encoder? How many ticks/revolution does that correspond to for the output shaft?

The resolution of the quad is 64 CPR, with the 30:1 gear the output resolution is 1920 CPR

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Topic: question to consider

The DC motor is polarized. Do we want to connect the negative (-) pole of the motor to the common ground (hint: look at the diagram above)? What happens if you plug the motor in opposite to the orientation shown in the diagram?

It will turn in the opposite direction.

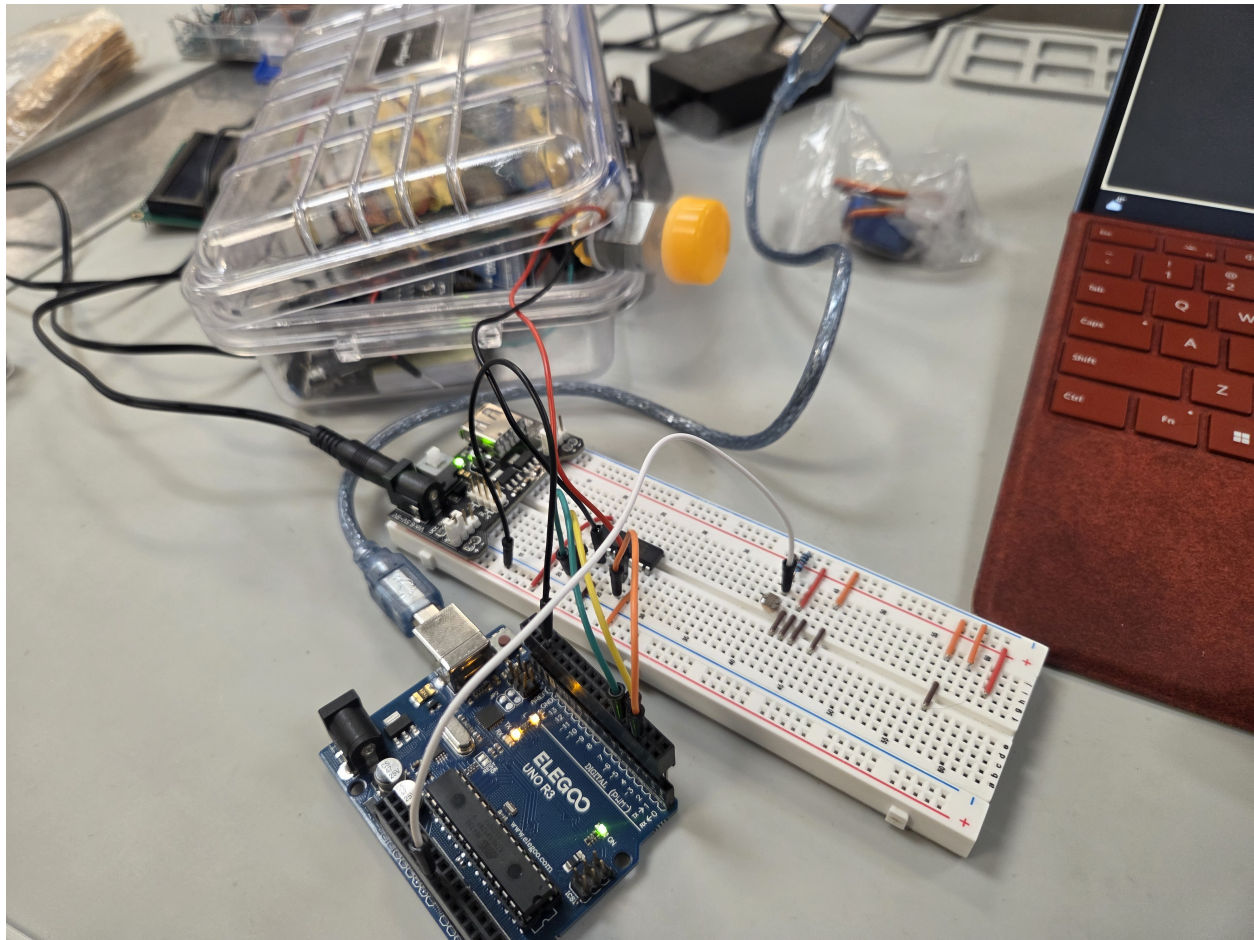


Figure 1: You want a brushed motor for a fan so that you have continuity through the whole range of motion while it spins.

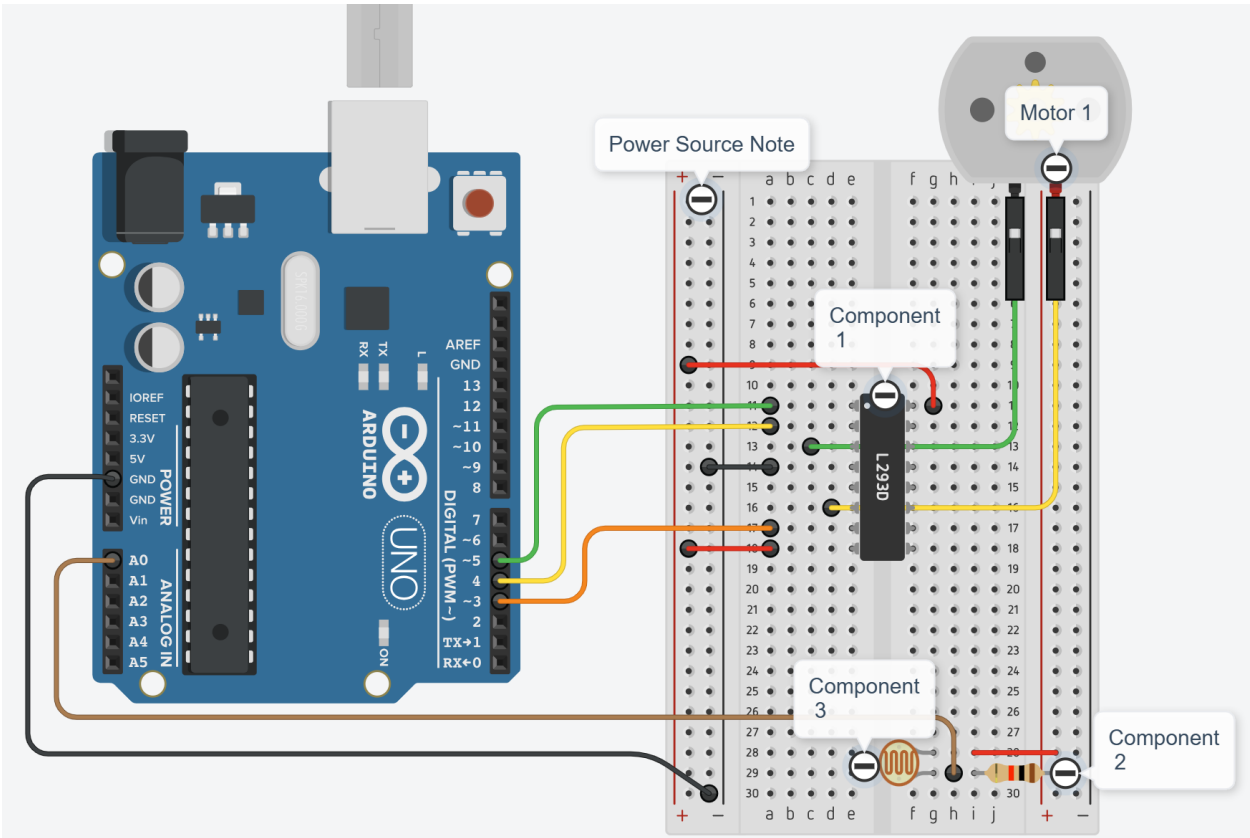


Figure 2: Tinker of 5.3

Name	Detail
PowerSource Note	Power Supply Module Connected
Component 1	L293D H Bridge
Component 2	$R_1 = 10k$
Component 3	Photo Sensor
Motor 1	DC Motor

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Topic: question to consider

Information about using PWM outputs with an Arduino Uno can be found here: <https://www.arduino.cc/en/t>
What range of values can you write for PWM? What range of values do you read from the photocell?

the range of PWM is 0-255, the photosensor reads off from 0 - 1023

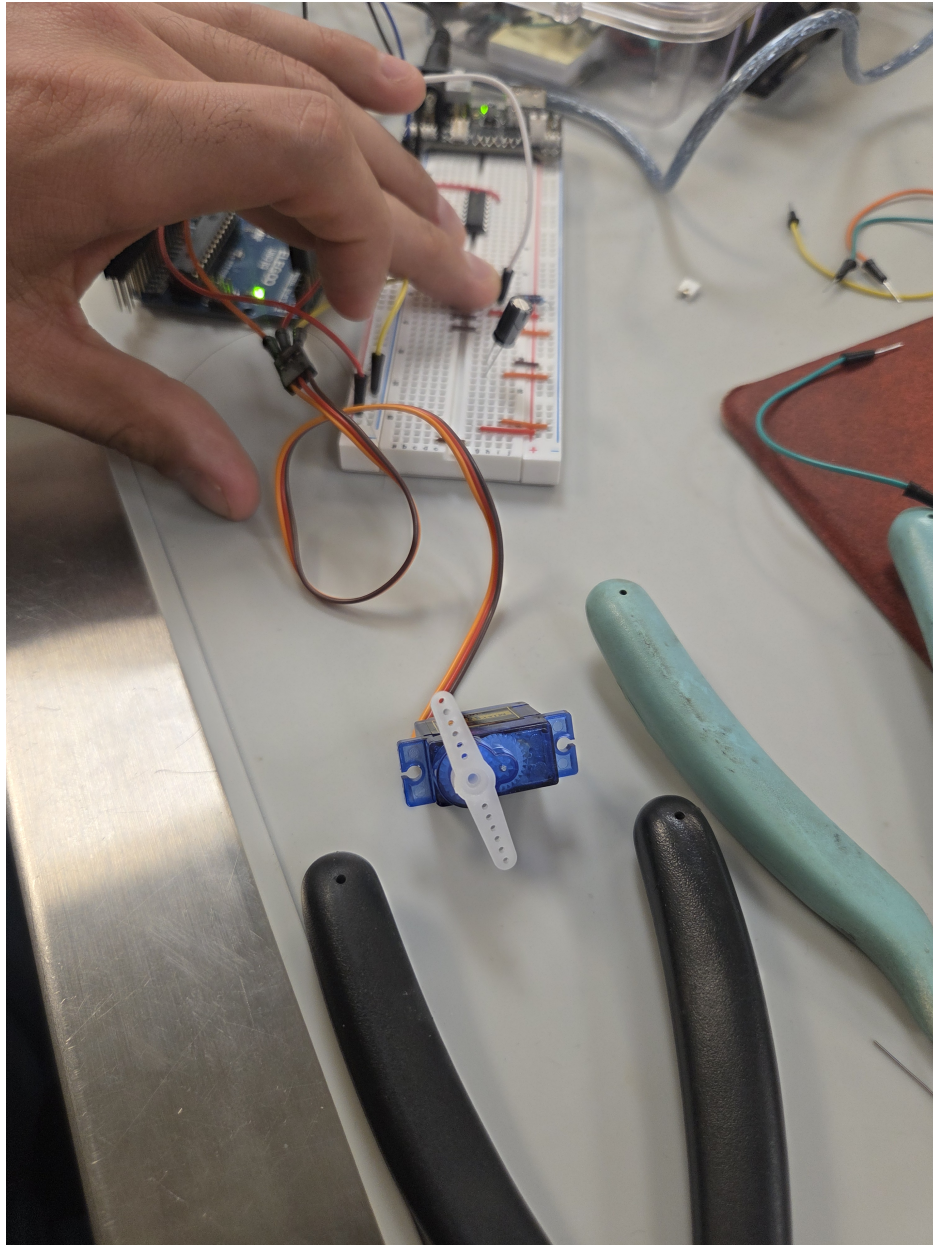


Figure 3: Poistion 1

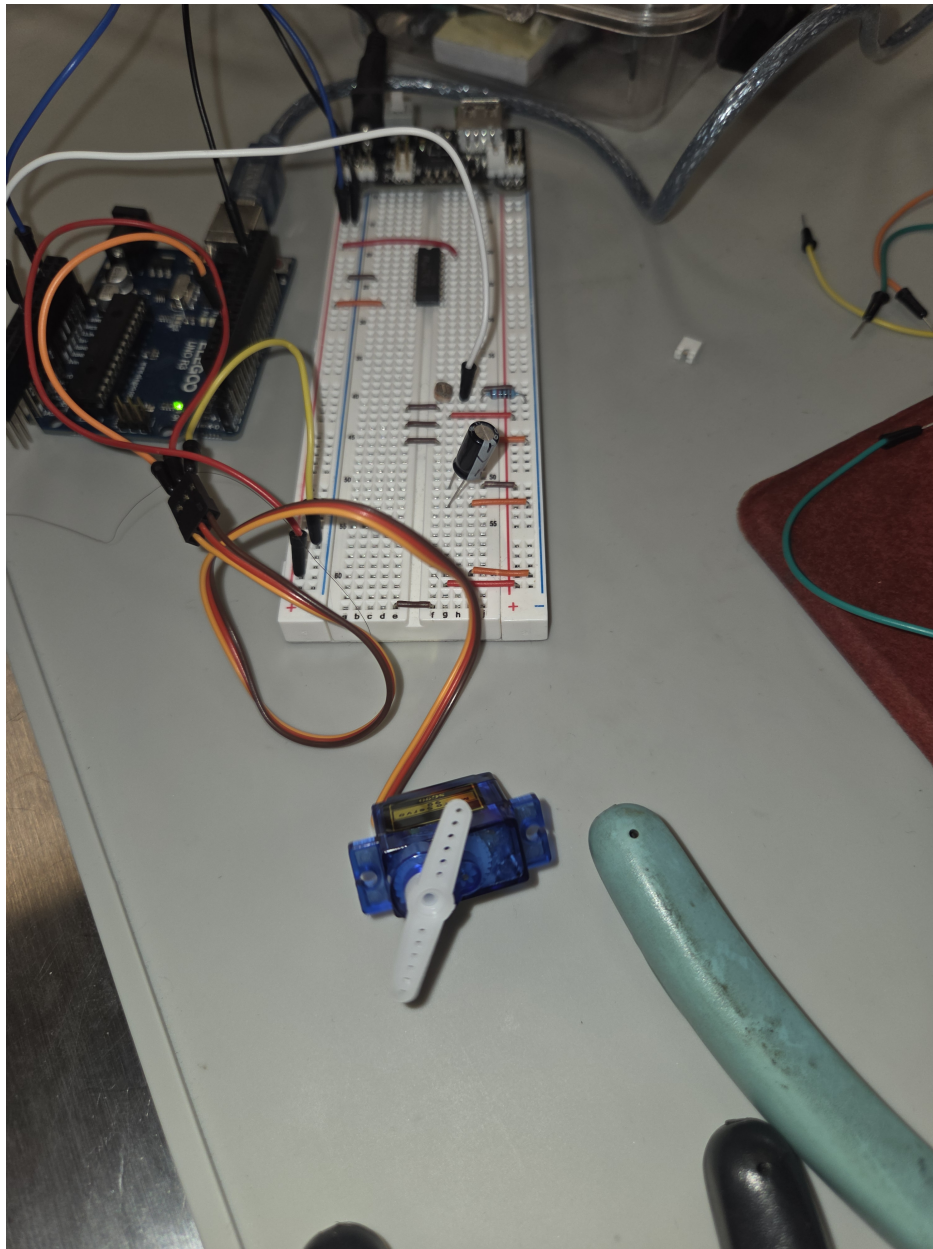


Figure 4: Position 2

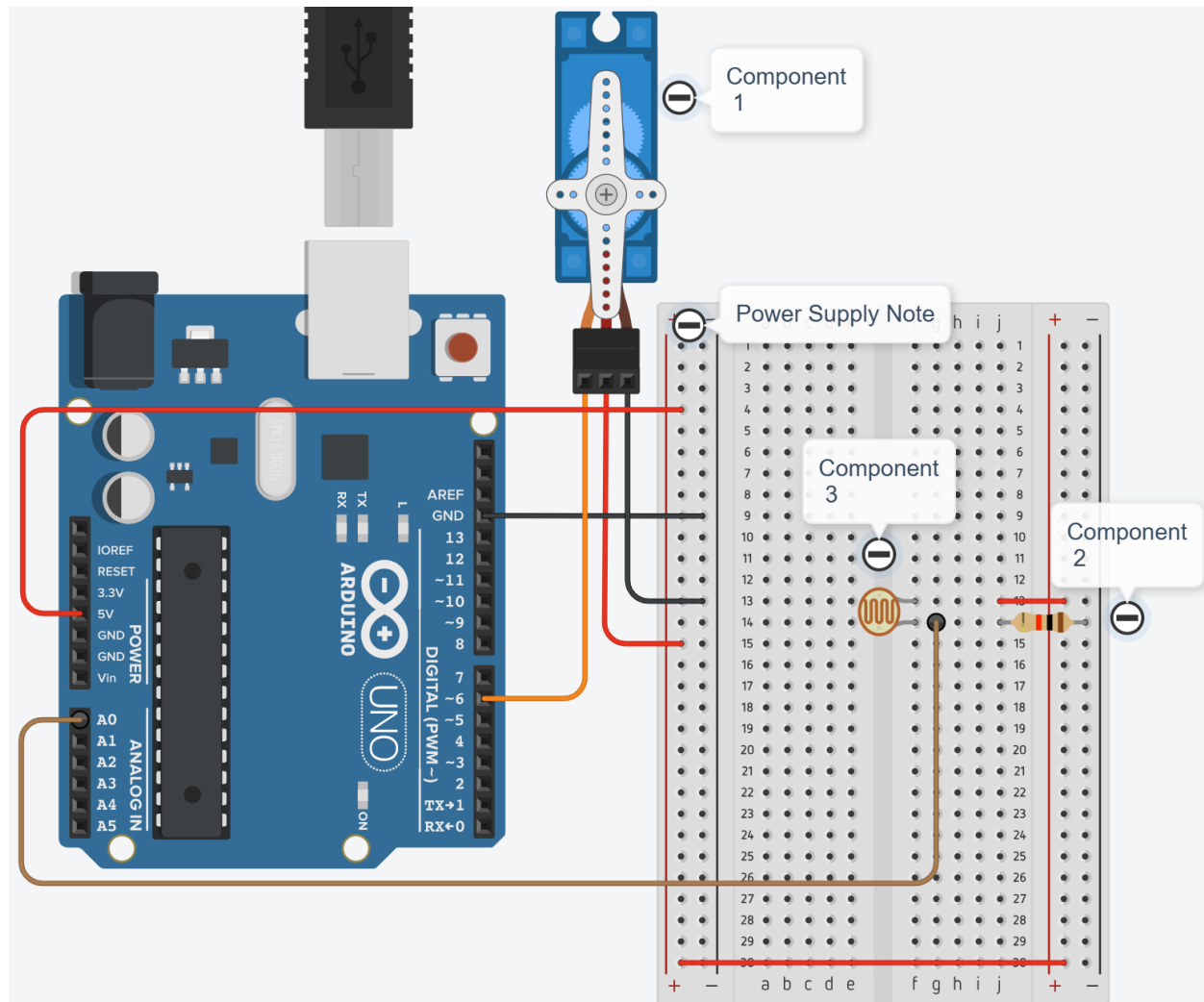


Figure 5: Tinker of 5.4

Name	Detail
PowerSource Note	Disabled Power Supply Module
Component 1	Servo
Component 2	$R_1 = 10k$
Component 3	Photo Sensor
Motor 1	DC Motor

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Topic: Question to consider

Hypothetically, let's pretend `analogRead()` did have a 50Hz PWM frequency (it's actually 490Hz or 980Hz by default, and in reality we'll end up using the Servo library to help with this process). What values would you want to write via `analogWrite()` to command the servo to move to positions -90, 0, and 90?

I would need to use 13, 19 and 25

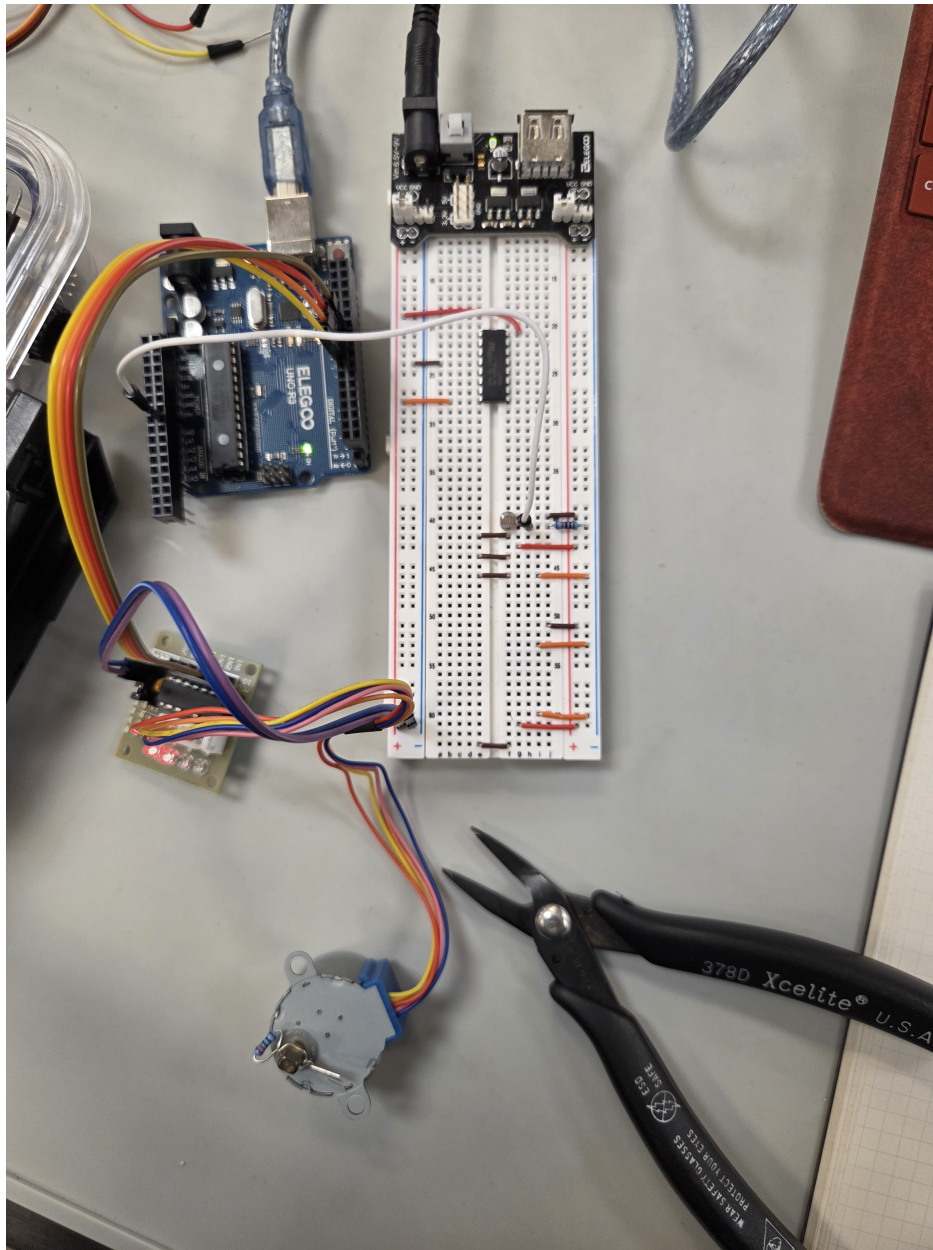


Figure 6: Completed stepper motor

Topic: Question to Consider

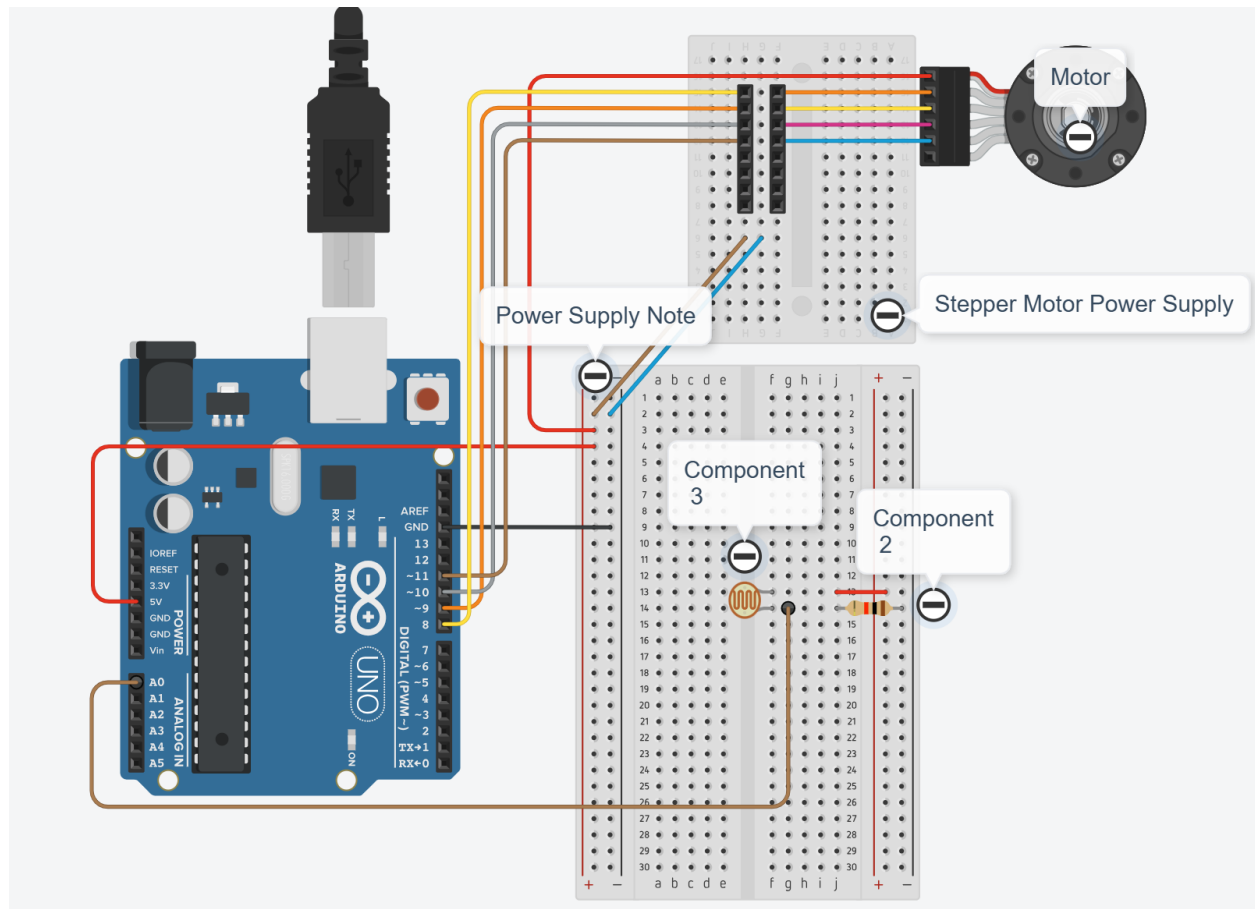


Figure 7: tinkr of 5.5

Name	Detail
PowerSource Note	Power Supply Module Connected
Stepper Motor Power Supply	attempt at recreating(Refer spec sheet)
Component 2	$R_1 = 10k$
Component 3	Photo Sensor
Motor 1	Stepper Motor

1. Motor Characterization Come up with a theoretical test to generate a torque-speed curve.

Why do we care about the torque-speed curve of a motor?

You do not have to actually run the test, just outline how it should be performed.

A useful resource can be found here: <https://web.archive.org/web/20210422011541/http://lancet.mit.edu/motors/motors3.html>.

the link provided didnt work but I used this video

<https://www.youtube.com/watch?v=UGXWl6Qhgzy>

I figured I could: Mount a small pulley on the motor shaft with radius r and positions S_1 & S_2 .

Wrap a belt around it.

Attach Tight-side and Slack-side to different springs.

Add an adjustable friction element so you can increase load.

Add tachometer to read RPM.

I could then measure current and voltage.

From here I would be able to find τ , ω , P_{mech} and $P_{electrical}$.

I would then just plot it all.

$$\tau = (S_1 - S_2)r_1$$

$$P_{mech} = \tau\omega$$

$$P = VI$$

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List online sources and collaborators.

no additional online sources outside of the lab links. I had no help from other students.

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Self-evaluation:

- (a) What did you expect to learn? *what a servo is how to use them in the code and the limits are.*
- (b) Rate class participation and needed adjustments. *This week was great. I started early and had it done on time.*
- (c) Grade yourself (0–4) and explain. *I would give myself a 4. Im happy with what I did.*
- (d) What did you actually learn? *I am getting a better understanding of how the gearing works and why we have them in servos and step motors.*

Code

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